magneto

_travelers guide _for mastering magnetism

4 playback heads | **3** modes | **2** min of tape

`[ tap ]` press `[ tap ]` hold › then noon = 12 o'clock cw / ccw = clockwise / counter

four playback heads, one record head · echo, looper, phrase sampler, spring reverb, oscillator

magneto · 28 hp · manual rev b

cyan = hold

01 buttons · LEDs

mg_01

`[ modes ]`	echo ● › loop ● › sample ● › echo …
`[ tap ]` ×2	delay time at head 4, 50 ms–15 s · loop / sample: splice in › out › clear (04)
`[ 1 ]`-`[ 4 ]`	● green = that head feeds back into the record head, even with its level at zero
`[ transport ]`	the four buttons turn ● red and become ∞ ◂▸ ▸‖ ‖ (06)

02 knobs

mg_02

`[ speed / pitch ]`	tape speed, 8:1 · min ¼ · about 9 o'clock ½ · noon normal · max 2× (fig.04)
`[ dry ]`	your input to the outputs
`[ wet ]`	all four heads to the outputs, after their levels
`[ rec lvl ]`	into the record head · clean to fat and saturated
`[ repeats ]`	feedback for every head whose button is green
`[ level 1–4 ]`	each head's volume · `[ ● n ]` + turn its knob = pan it (pan on center)

03 tape mechanics

mg_03

`[ low cut ]`	min = full lows · max = thin, peaky "magnetic drum" repeats
`[ tape age ]`	min = fresh tape · cw = warmer · max = a peak that blooms at high repeats
`[ crinkle ]`	friction, creases, splices, dirt · min clean · max chewed
`[ wow & flutter ]`	ccw = a tuned machine · cw = one that needs service
`[ spring ]`	spring reverb level · dry and wet both feed it

crinkle and wow & flutter bite much harder at slow speeds

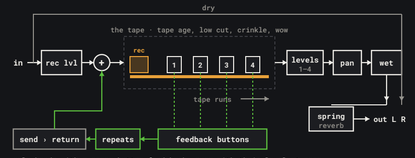
05 switches

mg_05

`[ heads ]`	**even** 1/16 · 1/8 · dotted 1/8 · 1/4 · **triplet** 1/16t · 1/8t · 1/4t · 1/4 · **shift** pitched heads (fig.02)
`[ pan ]`	**LRLR** 1 3 left · 2 4 right · **center** all middle, custom pans · **LRRL** 1 4 left
one output	everything sums to mono at that jack

__fig.01 : signal path

mono in, stereo out



only heads with a green button feed back, even with their level at zero
loop / sample: head 4 sits before the record head and plays the loop

04 modes

mg_04 · `[ modes ]`

● echo	**four-head tape delay** · `[ tap ]` sets the head spacing, `[ speed ]` the tape speed
● loop	**sound-on-sound** · head 4 plays the loop, heads 1–3 keep echoing · needs `[ 4 ]` green and level 4 up
loop `[ tap ]`	splice in › splice out (plays) › clear · `[ repeats ]` lower = the loop fades as you layer
● sample	**phrase sampler** · `[ tap ]` record › stop › clear · `[ ▸| ]` plays it once
length	longest loop or sample: 15 s at max speed · 30 s noon · 1 min half · 2 min min
heads 1–3	keep the delay time you had when you entered loop / sample, or follow `[ clk in ]`

__fig.04 : the speed dial

8:1



slower
longer, lower, darker · wow and crinkle grow

faster
shorter, higher, cleaner

tap instead
moves the heads: new delay, same pitch

turning it
repitches what's on the tape

09 audio

mg_09 · 20 vpp

mono in	`[ left in ]` only
mono out	`[ left out ]` only · all sums to it
stereo out	both outs · works from a mono in too
`[ send ]` `[ return ]`	fx loop before the record head · both patched = repeats go mono
the module	28 hp · 41 mm · 210 mA each rail · 96 khz

__fig.02 : head spacing

and the clk outs



tapped or clocked at 1000 ms · clk 1–4 out tick at these times, 50% duty, phase-aligned
shift = each head plays at its own speed · loop / sample: clk 4 out = the loop length

07 pitch quantize

mg_07 · `[ tap ]`

`[ tap ]`	`[ speed / pitch ]` snaps to the notes of a scale · tap LED ●
`[ transport ]` + `[ speed ]`	turn to pick one of 15 scales · buttons 1–4 show which (fig.05)
range	two octaves (4:1) · chromatic 1.2 octaves · speed cv snaps to half steps

__fig.05 : the 15 scales

`[ transport ]` + `[ speed ]`



08 self-oscillation

mg_08 · no input needed

1. `[ speed / pitch ]` fully ccw · `[ repeats ]` at 100%
2. `[ tap ]` ×2, as fast as you can · it starts to sing
3. play `[ speed ]` and the four tape knobs · it wanders through pitches
4. `[ transport ]` › `[ ∞ ]` locks the pitch · speed and speed cv still play it

speed cv is 1 v/oct, −3 to +3 v: six octaves, three below the knob alone · LRLR / LRRL widen the tone

12 dip switches

mg_12 · on the back, off by default

s1 `[ feedback cv ]`	on: with the transport LED off, the transport jacks toggle feedback buttons 1–4 instead
s2 `[ dual split ]`	on: left in / out = mono four-head delay · right in / out = mono spring reverb

06 transport

mg_06 · `[ transport ]` first

`[ ∞ ]` infinite	record head off · the last delay cycle or loop plays forever
`[ ◂▸ ]` reverse	from that moment · echo: the whole tape backwards · loop / sample: the loop backwards
`[ ▸| ]` restart	loop / sample from the top · realigns shift heads · leaves pause instantly
`[ ‖ ]` pause	the tape slows to a stop, or spins back up · while paused, level 4 = how fast
scrub	while paused, `[ speed / pitch ]` scrubs about a foot of tape · slower speed = more audio
defaults	all latch except ▸|, always a trigger · change one at power on (mg_15)

10 cv inputs

mg_10

continuous, −5 to +5 v — the knob is the offset

`[ speed ]`	−3 to +3 v, 1 v/oct
`[ wet ]` `[ repeats ]` `[ spring ]`	add to their knobs

triggers, 0–5 v rising edge

`[ clk in ]`	delay time as a quarter note, 50 ms–15 s
`[ rec gate ]`	record head on / off · rec lvl LED ● when muted
`[ shift ]`	shift head mode on / off
`[ ∞ ]` `[ ◂▸ ]` `[ ▸| ]` `[ ‖ ]`	the transport buttons
`[ tap ]`	the tap button: delay, splice or sample · 10 ms–15 s

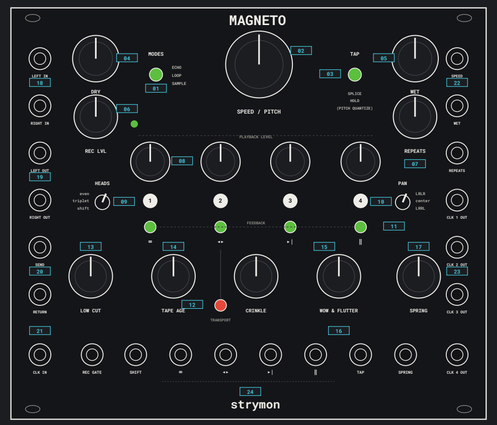
11 tricks

mg_11 · from the manual

sub-octave	shift heads · a gate into `[ ▸| ]` cv = a zero-latency sub-octave from head 3
endless	sample mode · `[ clk 4 out ]` › `[ ▸| ]` cv retriggers the phrase
extra tone	`[ ∞ ]` on a very short delay adds a harmonic at the seam

__fig.03 : front panel

01–17 = controls · 18–24 = jacks



[01]	__modes : echo · loop · sample	[13]	__low cut : thins the repeats
[02]	__speed / pitch : tape speed 8:1	[14]	__tape age : darker, peakier
[03]	__tap : delay · splice · hold = quantize	[15]	__crinkle : damaged tape
[04]	__dry : input level	[16]	__wow & flutter : speed wobble
[05]	__wet : all heads	[17]	__spring : reverb level
[06]	__rec lvl : into the tape · LED	[18]	__ins : left = mono
[07]	__repeats : feedback	[19]	__outs : one = mono sum
[08]	__level 1–4 : each head	[20]	__send · return : before record
[09]	__heads : even · triplet · shift	[21]	__clk in : quarter note
[10]	__pan : LRLR · center · LRRL	[22]	__cv : speed · wet · repeats
[11]	__1–4 : feedback · transport	[23]	__clk 1–4 out : the heads
[12]	__transport : flips 11 to red	[24]	__gates : rec · shift · transport · tap

_where things hide

no menus · mg_15

held buttons

- `tap` pitch quantize
- `transport + speed` pick a scale
- `green n + level n` pan head n (pan center)
- `paused: level 4` slowdown speed

at power on

- `hold ∞ ◂▸ ▸| or ‖` flips it: momentary ● / latch ●
- `dip switches` s1 feedback cv · s2 dual split (12)

LED codes

`modes · tap`	● echo ● loop ● sample ● tap = quantize
`buttons 1–4`	● feedback on ● transport layer / scale code
`rec lvl`	● none ● clean ● saturating ● heavy